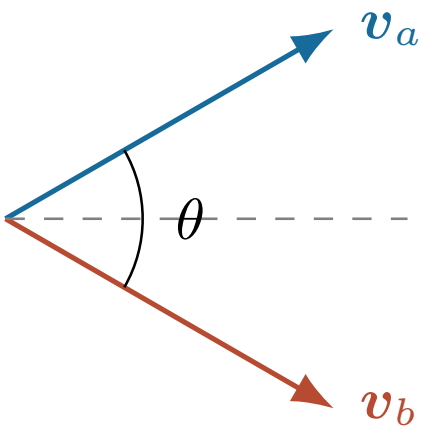
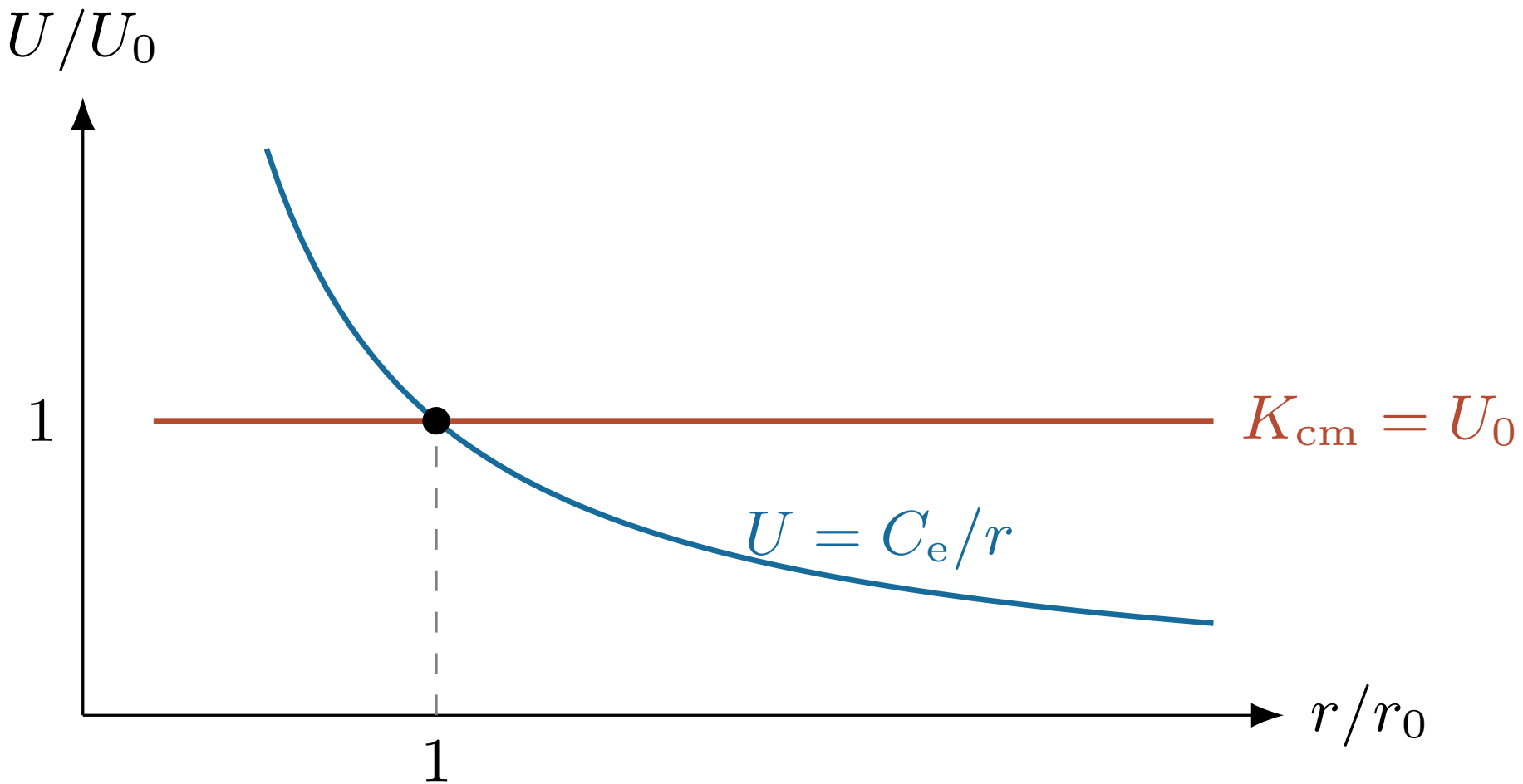
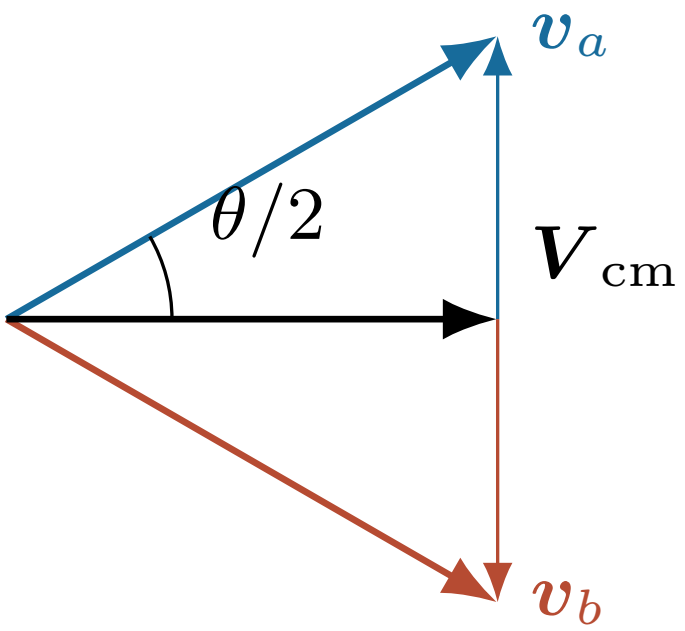
等速正碰



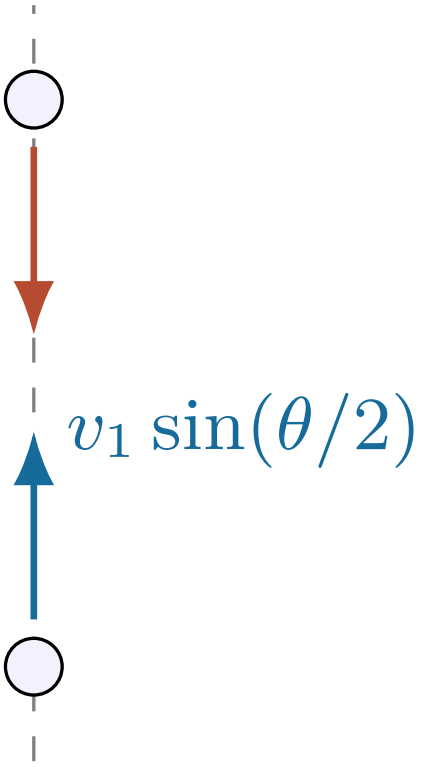
夹角速度图，不表示初始位置



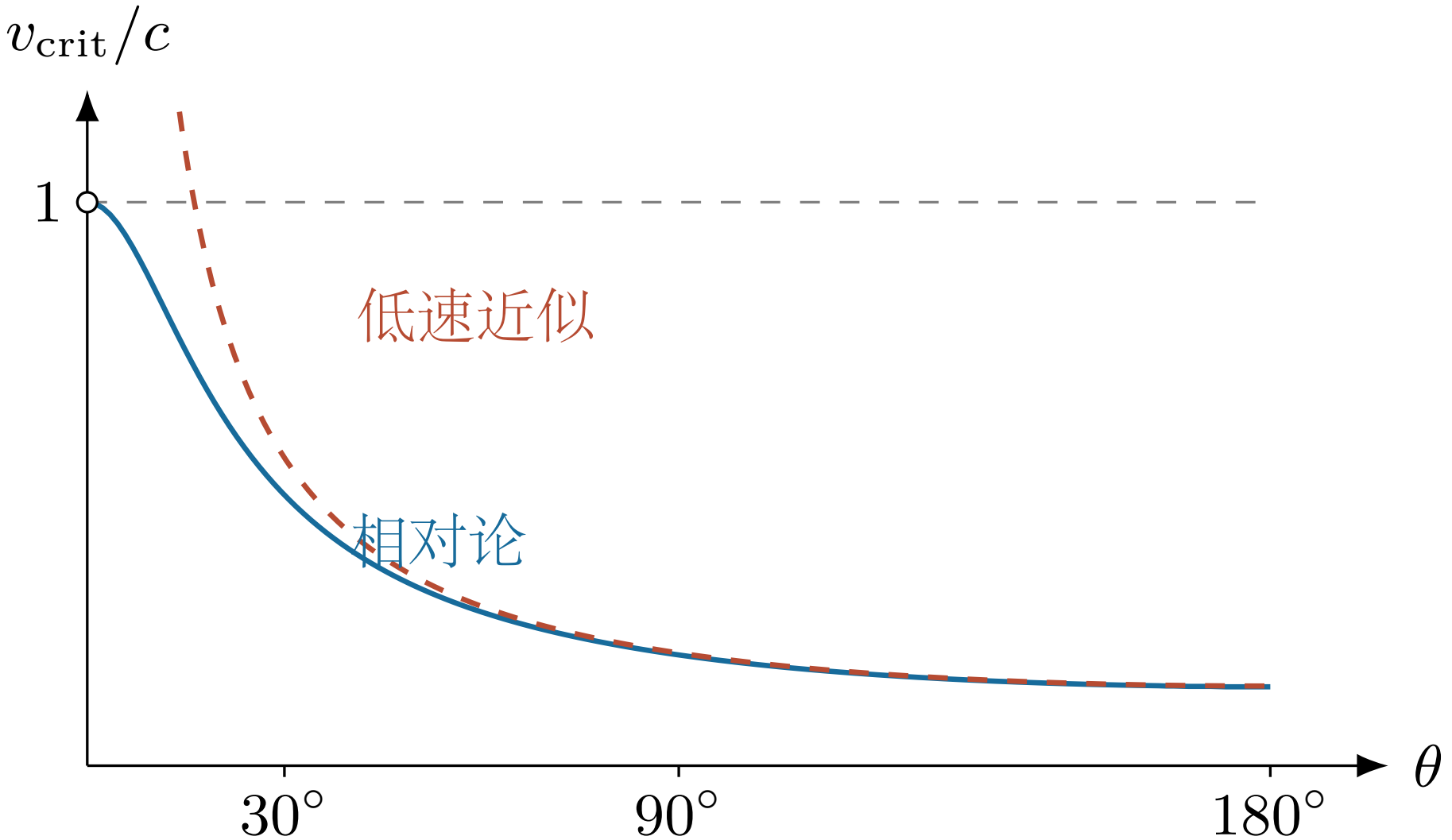
临界最近距离 $r_{\min} = r_0$



非相对论分量分解



质心系反向运动



示例 $\delta = U_0/(m_0c^2) = 0.02$